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MLA call

Matteo Scialpi^{1,2}, Francesco Di
Clemente³, Leigh Smith^{4,5}, and Michał
Bejger^{2,6}

APRIL: Auxiliary Physically-Redundant Information in Loss

A physics-informed framework for parameter estimation with a gravitational-wave case study

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¹Dipartimento di Fisica e Scienze della Terra, Università di Ferrara, Via Saragat 1, 44122 Ferrara, Italy. ²INFN Sezione di Ferrara, Via Saragat 1, 44122 Ferrara, Italy.

³Department of Physics, University of Houston, Houston, TX 77204, USA. ⁴Dipartimento di Fisica, Università di Trieste, Via Valerio 2, 34127 Trieste, Italy. ⁵INFN, Sezione di Trieste, I-34127 Trieste, Italy.

⁶Nicolaus Copernicus Astronomical Center, Polish Academy of Sciences, Bartycka 18, 00-716 Warszawa, Poland.



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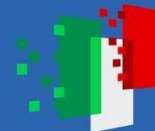
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- Introduction
- Theoretical framework
- Loss benchmark applied to GW parameter estimation
- Conclusions and future steps



Motivation 1 – PE for unmodeled searches

- Unmodeled (or minimally-modeled) searches are mostly based on coherent excess power analysis.
 - Pro: new signals can be detected.
 - Con: when dealing with CBC, PE on the source properties is done only with a fit on the reconstructed waveform.
- We can solve the Con thanks to Physics-Informed Neural Networks (PINNs).
- APRIL doesn't solve the PE problem in its full form, but can contribute as the core of a future PE algorithm.



Motivation 2 - PINNs lacks in solutions for multi-system inverse problems

- Most used Physics-Informed Neural Networks (PINNs [2]) are based on the solution of partial differential equations describing **single realizations** of systems (ex. Navier-Stokes in fluid dynamics).
- Inverse problems are possible, but **for more than one or few realizations** per time **the computational cost will become unmanageable** [3].



Machine learning with auxiliary physically-redundant information included in loss for gravitational waves parameter estimation

Abstract

Physics-Informed Neural Networks (PINNs) embed the partial differential equations (PDEs) governing the system under study directly into the training of Neural Networks, ensuring solutions that respect physical laws. While effective for single-system problems, standard PINNs scale poorly to datasets containing many realizations of the same underlying physics with varying parameters. To address this limitation, we present a complementary approach by including auxiliary physically-redundant information in loss (APRIL), i.e. augment the standard supervised output-target loss with auxiliary terms which exploit exact physical redundancy relations among outputs. We mathematically demonstrate that these terms preserve the true physical minimum while reshaping the loss landscape, improving convergence toward physically consistent solutions. As a proof-of-concept, we benchmark APRIL on a fully-connected neural network for gravitational wave (GW) parameter estimation (PE). We use simulated, noise-free compact binary coalescence (CBC) signals, focusing on inspiral-frequency waveforms to recover the chirp mass $\mathcal{M}$, the total mass M_{tot} , and symmetric mass ratio η of the binary. In this controlled setting, we show that APRIL achieves up to an order-of-magnitude improvement in test accuracy, especially for parameters that are otherwise difficult to learn. This method provides physically consistent learning for large multi-system datasets and is well suited for future GW analyses involving realistic noise and broader parameter ranges.



Theoretical framework

- Given a dataset $\mathbf{x}$ and a set of features $\mathbf{y}$, the NN aims to approximate $\hat{\mathbf{y}}_i = \mathcal{F}(\mathbf{x}_i)$. To do so, an optimization on the NN parameters is performed minimizing a loss function.

$$\mathcal{L}_t(\theta) = \text{MSE}(\mathbf{y}_{\theta,i}, \hat{\mathbf{y}}_i) = \frac{1}{B} \sum_{i=1}^B (\mathbf{y}_{\theta,i} - \hat{\mathbf{y}}_i)^2$$

- True relation among $\mathbf{y}$ features:

$$\hat{y}_{1,i} = g(\hat{y}_{2,i}, \hat{y}_{3,i}, \dots, \hat{y}_{n,i})$$

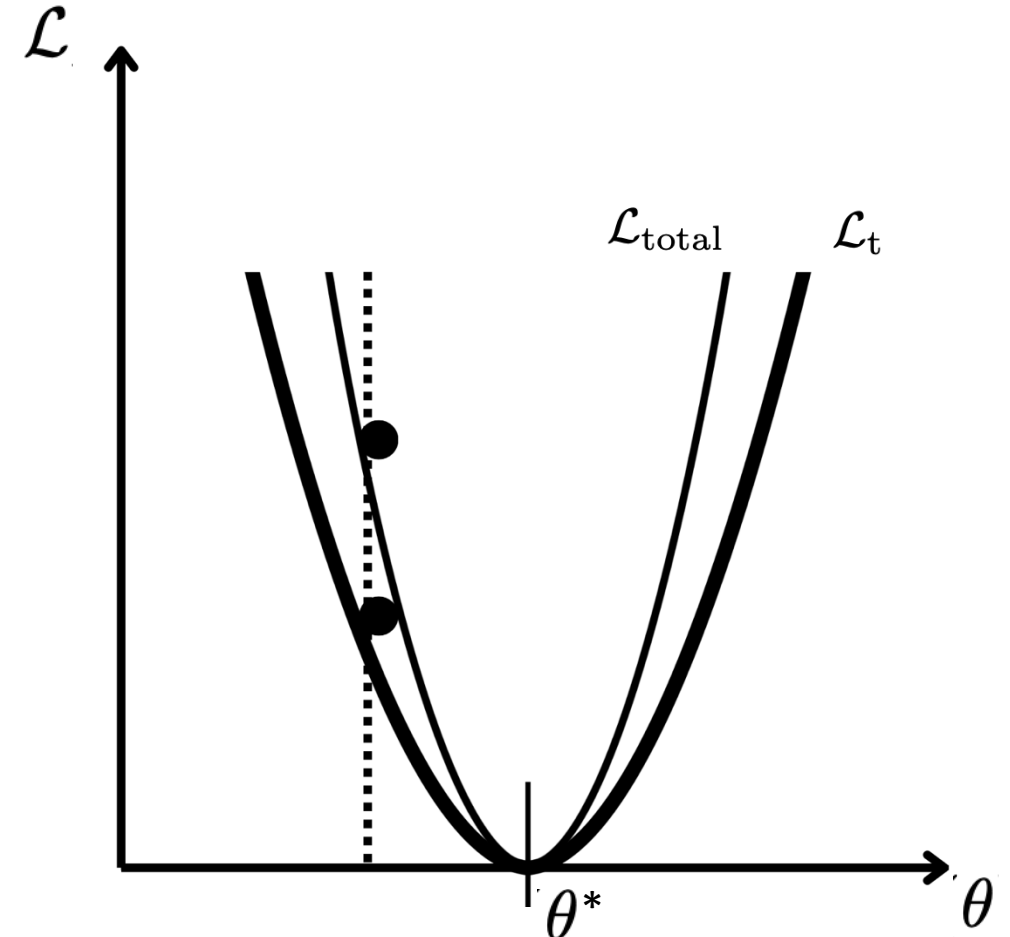
- New loss term **highlighting the redundancy between output features**:

$$\mathcal{L}_{\text{APRIL}}(\theta) = \text{MSE}(g(y_{\theta,2,i}, y_{\theta,3,i}, \dots, y_{\theta,n,i}), y_{\theta,1,i})$$

- New loss:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_t(\theta) + \mathcal{L}_{\text{APRIL}}(\theta)$$

- Adding new redundant (physical) term we are sharpening the global minimum in the θ direction.





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Loss benchmark applied to GW PE

We **benchmark different loss terms** on a simple fully connected neural network to systematically test this new technique.



Training dataset

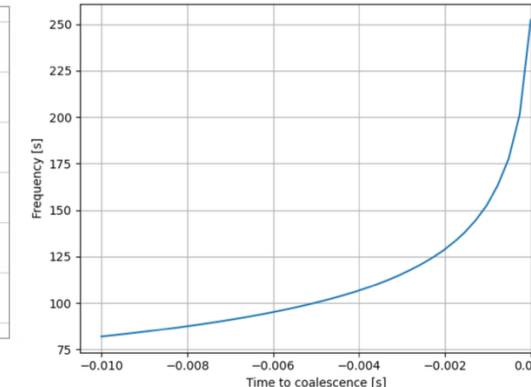
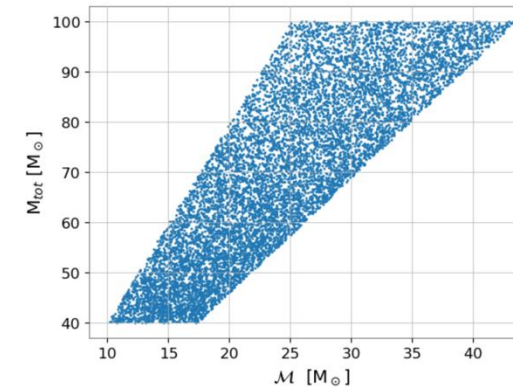
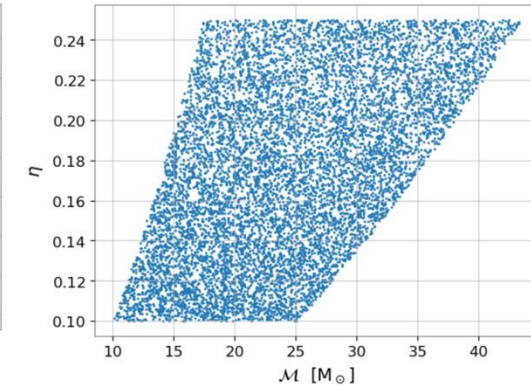
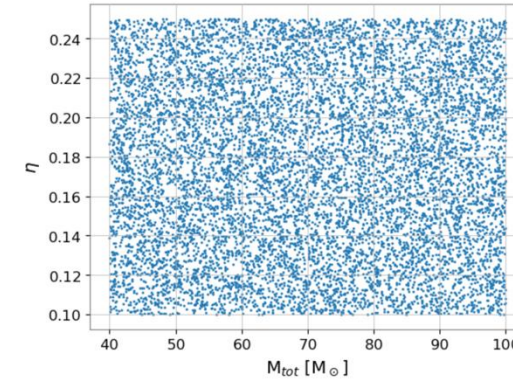
- Uniform sampling:
 - $M_{\text{tot}} \in [40, 100] M_{\odot}$
 - $\eta \in [0.1, 0.25]$
- M_{chirp} , m_1 and m_2 are calculated from M_{tot} and η and stored as ground truth.

- $\frac{df}{dt}$ is calculated up to 1.5PN correction:

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_{\odot}}{c^3} \frac{\mathcal{M}}{M_{\odot}} \right)^{5/3} f^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) \varepsilon + 4\pi \varepsilon^{3/2} \right]$$

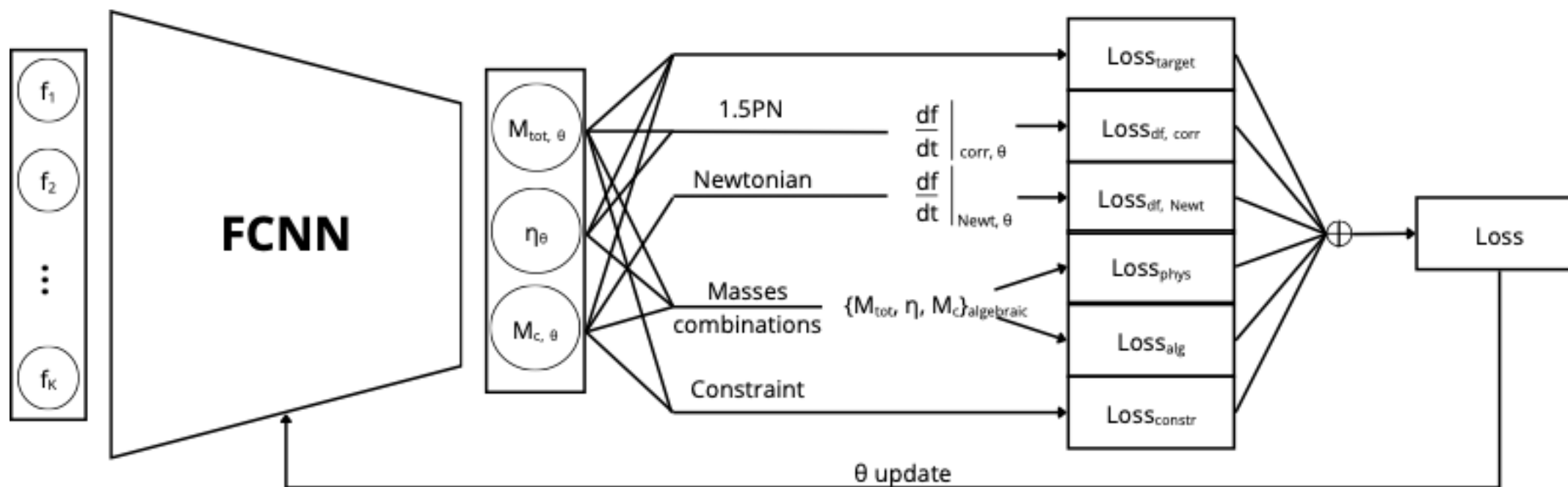
$$\varepsilon = \left(\frac{GM_{\odot}}{c^3} \frac{M_{\text{tot}}}{M_{\odot}} \pi f \right)^{2/3}$$

- $\frac{df}{dt}$ is integrated to have $\mathbf{f}(\mathbf{t}_k)$ array.





Architecture





Loss terms

$$\mathcal{L}_{\text{total}}(\theta) = \alpha_t \mathcal{L}_t(\theta) + \alpha_{\text{APRIL}} [\mathcal{L}_{\text{df}}(\theta) + \mathcal{L}_p(\theta) + \mathcal{L}_a(\theta)] ,$$

$$\begin{aligned} \mathcal{L}_t(\theta) &= \text{MSE}(\mathcal{M}_\theta, \hat{\mathcal{M}}) + \text{MSE}(M_{\text{tot}, \theta}, \hat{M}_{\text{tot}}) + \text{MSE}(\eta_\theta, \hat{\eta}) . \\ \mathcal{L}_{\text{df}} &= \beta \mathcal{L}_{\text{Newt}} + \gamma \mathcal{L}_{\text{corr}} = \beta \text{MSE} \left(\left. \frac{df}{dt} \right|_{\text{Newt, alg}}, \left. \frac{df}{dt} \right|_{\text{Newt}}^\wedge \right) + \gamma \text{MSE} \left(\left. \frac{df}{dt} \right|_{\text{corr, alg}}, \left. \frac{df}{dt} \right|_{\text{corr}}^\wedge \right) \\ \mathcal{L}_p &= \text{MSE}(\mathcal{M}_{\text{alg}}, \mathcal{M}_\theta) + \text{MSE}(M_{\text{tot, alg}}, M_{\text{tot}, \theta}) + \text{MSE}(\eta_{\text{alg}}, \eta_\theta) \\ \mathcal{L}_a &= \text{MSE}(\mathcal{M}_{\text{alg}}, \hat{\mathcal{M}}) + \text{MSE}(M_{\text{tot, alg}}, \hat{M}_{\text{tot}}) + \text{MSE}(\eta_{\text{alg}}, \hat{\eta}) . \end{aligned}$$

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_\odot}{c^3} \frac{\mathcal{M}}{M_\odot} \right)^{5/3} f^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) \varepsilon + 4\pi \varepsilon^{3/2} \right] \quad \varepsilon = \left(\frac{GM_\odot}{c^3} \frac{M_{\text{tot}}}{M_\odot} \pi f \right)^{2/3}$$

$$\mathcal{M}_{\text{alg}} = M_{\text{tot}, \theta} \eta_\theta^{3/5}, \quad M_{\text{tot, alg}} = \frac{\mathcal{M}_\theta}{\eta_\theta^{3/5}}, \quad \eta_{\text{alg}} = \left(\frac{\mathcal{M}_\theta}{M_{\text{tot}, \theta}} \right)^{5/3} .$$



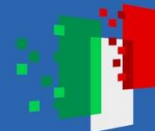
Benchmark study

$$\mathcal{L}_{\text{total}}(\theta) = \alpha_t \mathcal{L}_t(\theta) + \alpha_{\text{APRIL}} [\mathcal{L}_{\text{df}}(\theta) + \mathcal{L}_p(\theta) + \mathcal{L}_a(\theta)] ,$$

- 3 different cases: $\{\alpha_t, \alpha_{\text{APRIL}}\} = \{1,0\}, \{0,1\}, \{1,1\}$.
- Common test dataset, using as score metric

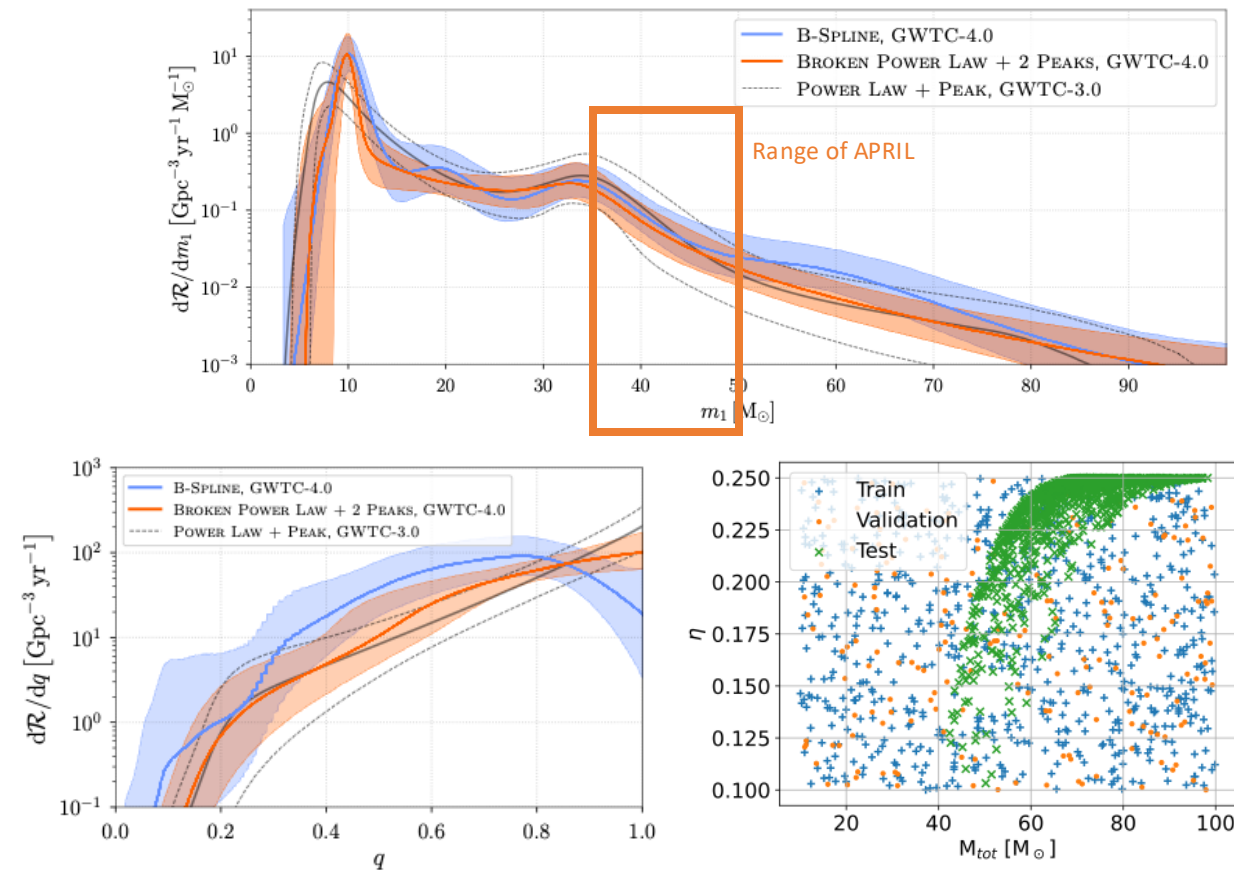
$$\text{RL1}(x_{\theta,i}, \hat{x}_i) = \frac{1}{T} \sum_{i=1}^T \frac{|x_{\theta,i} - \hat{x}_i|}{\hat{x}_i}$$

- The study was repeated for 5 different seeds, set before the training loop.
- Everything was performed varying the training data size D ($\{1,2,5\} \times 10^3$) and the batch size B ($\{16,32,64\}$).



Common test dataset: GWTC-4 inferred BBH distribution (Broken Power Law + 2 Peaks)

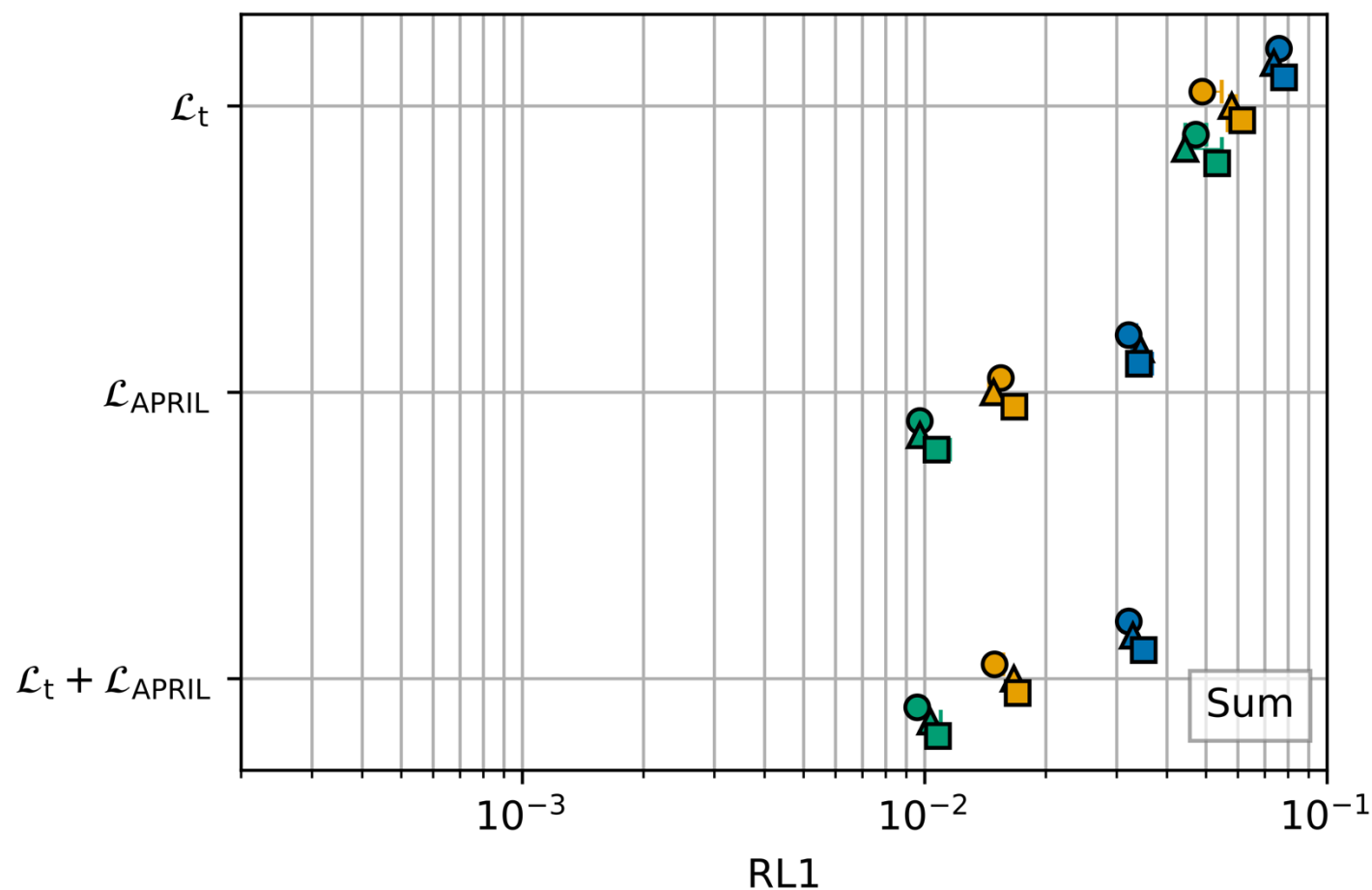
- We used **rejection sampling** from **GWTC-4 inferred BPL2P distribution** [4,5] to create a 10^3 points test dataset as similar as possible to the real case.
- To be consistent with the training dataset range, m_1 was sampled from 35 to $50 M_\odot$, while q from 0.1 to 1.
- The resulting f_k was given as point-like estimation (not through pytorch dataset), like will be done for a new incoming event.



[The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, "Gwtc-4.0: Population properties of merging compact binaries," 2025.]



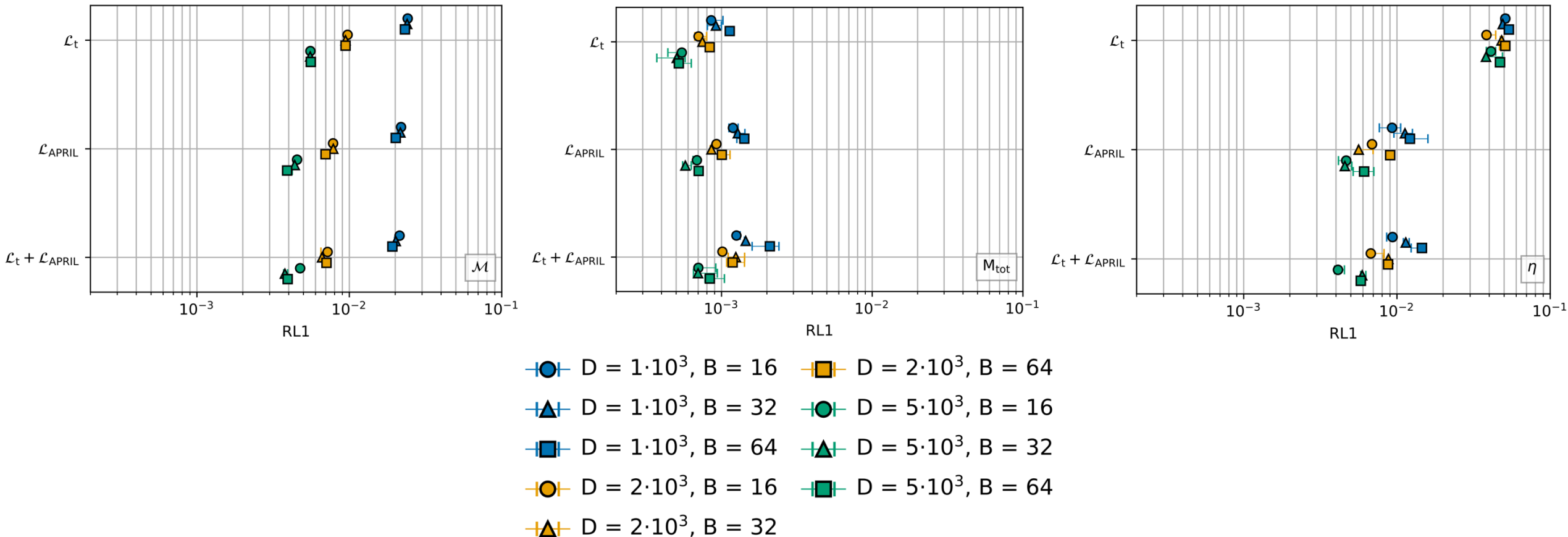
Results



- Blue circle: $D = 1 \cdot 10^3, B = 16$
- Blue triangle: $D = 1 \cdot 10^3, B = 32$
- Blue square: $D = 1 \cdot 10^3, B = 64$
- Orange circle: $D = 2 \cdot 10^3, B = 16$
- Orange triangle: $D = 2 \cdot 10^3, B = 32$
- Orange square: $D = 2 \cdot 10^3, B = 64$
- Green circle: $D = 5 \cdot 10^3, B = 16$
- Green triangle: $D = 5 \cdot 10^3, B = 32$
- Green square: $D = 5 \cdot 10^3, B = 64$



Results





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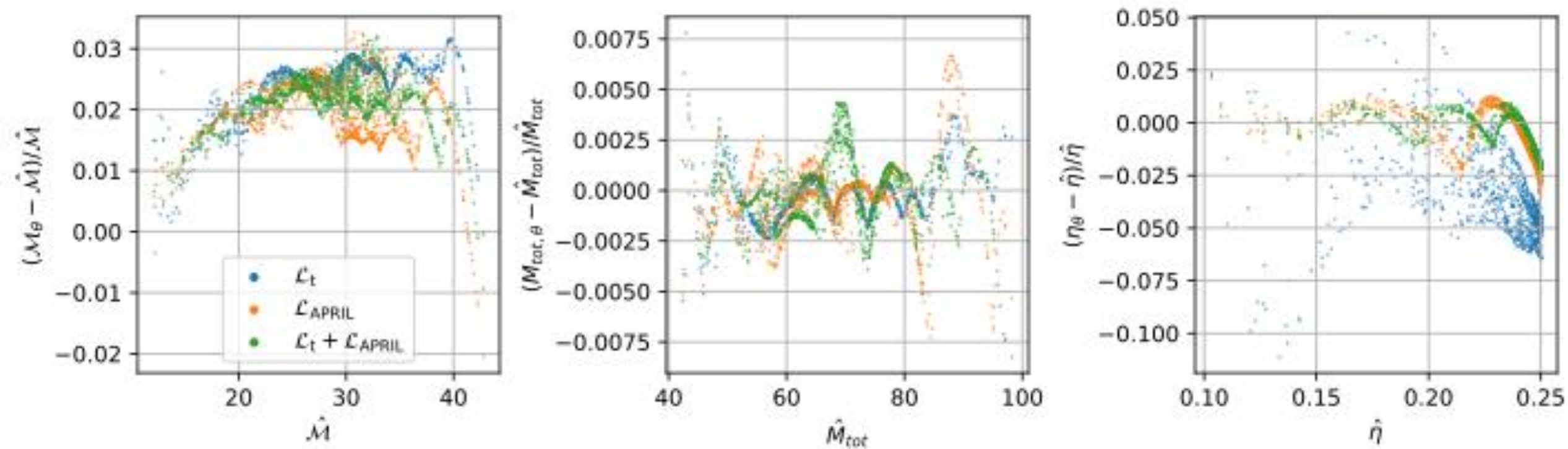
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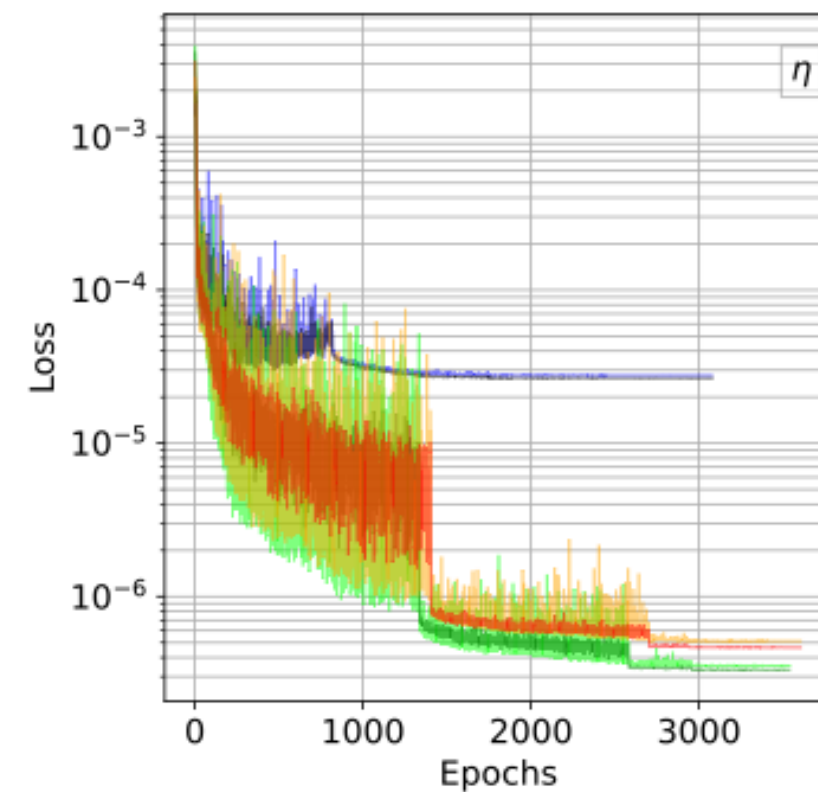
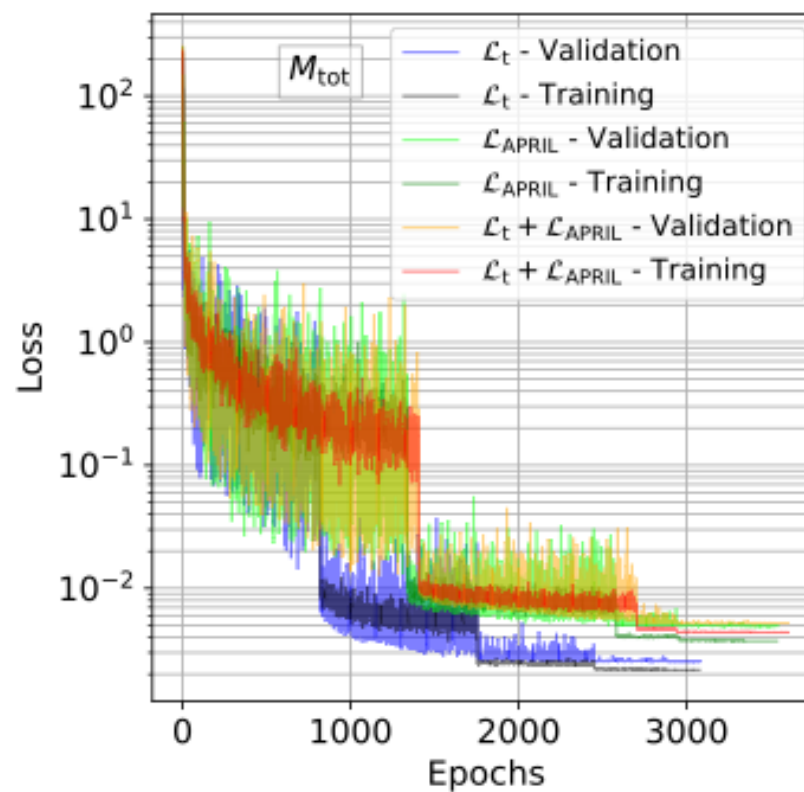
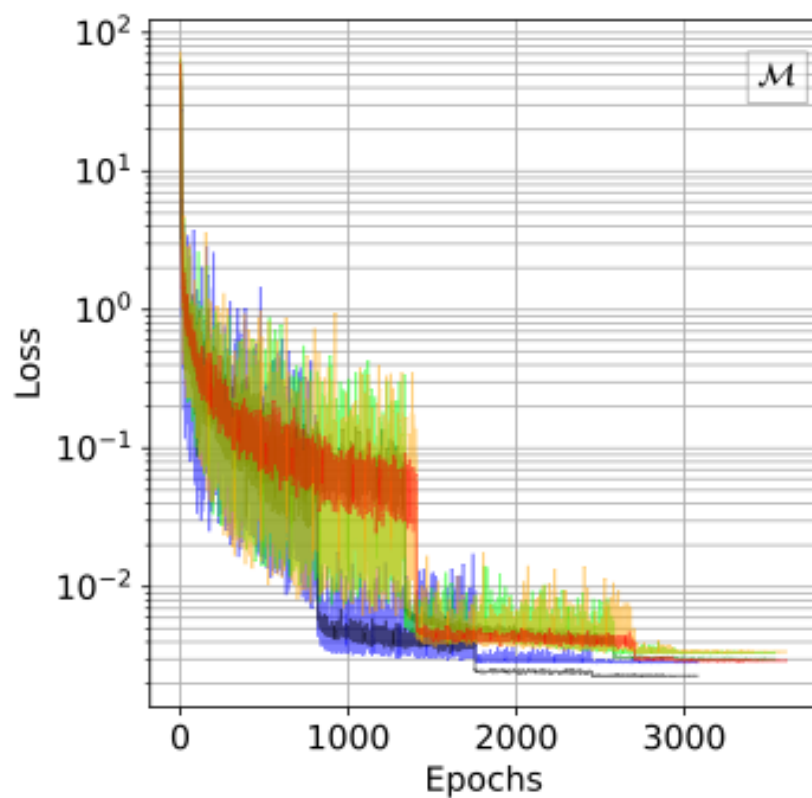


Results - Test dataset details



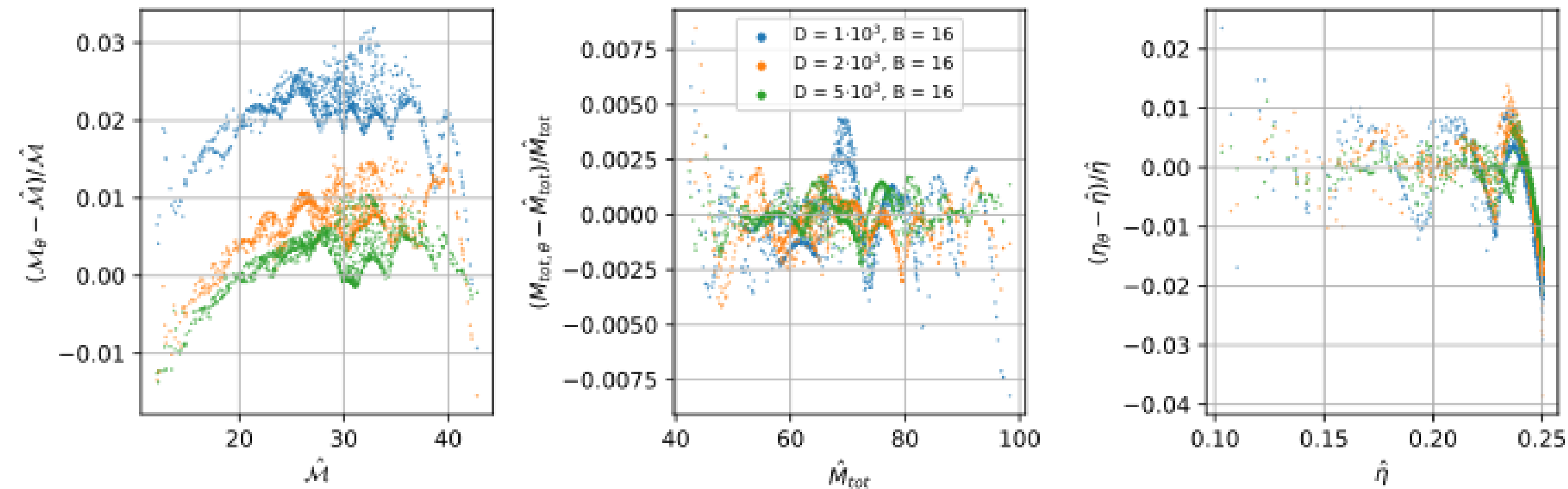


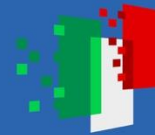
Results - Action inside the training





Results - Dependence on data size





Conclusions

- **Auxiliary Physics-Informed Loss terms are extremely helpful both for implementing physical relations inside a NN and as regularizers toward the global minimum of the loss, which is the only one with physical sense.**
- We demonstrated the action on the **sharpening of the local minimum.**
- We benchmarked APRIL with respect to the ground truth term for GW PE, **gaining a factor 10 in performance.**
- APRIL terms could be also suitable for a **large range of applications**, estimating parameters for a high number of realization of the same physical system.
- **Future steps:**
 - Increase to the full GWTC-4 mass range.
 - We are developing a NN for PE on CBC from cWB outputs, based on APRIL core.



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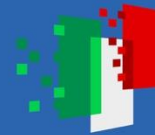
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Thank you for your attention



References

1. F. Di Clemente, M. Scialpi, and M. Beijer, “Explainable autoencoder for neutron star dense matter parameter estimation,” 2025.
2. M. Raissi, P. Perdikaris, and G. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
3. E. Torres, J. Schiefer, and M. Niepert, “Adaptive physics-informed neural networks: A survey,” 2025.
4. The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, “Gwtc-4.0: Updating the gravitational-wave transient catalog with observations from the first part of the fourth ligo-virgo-kagra observing run,” 2025.
5. The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, “Gwtc-4.0: Population properties of merging compact binaries,” 2025.



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Backup slides



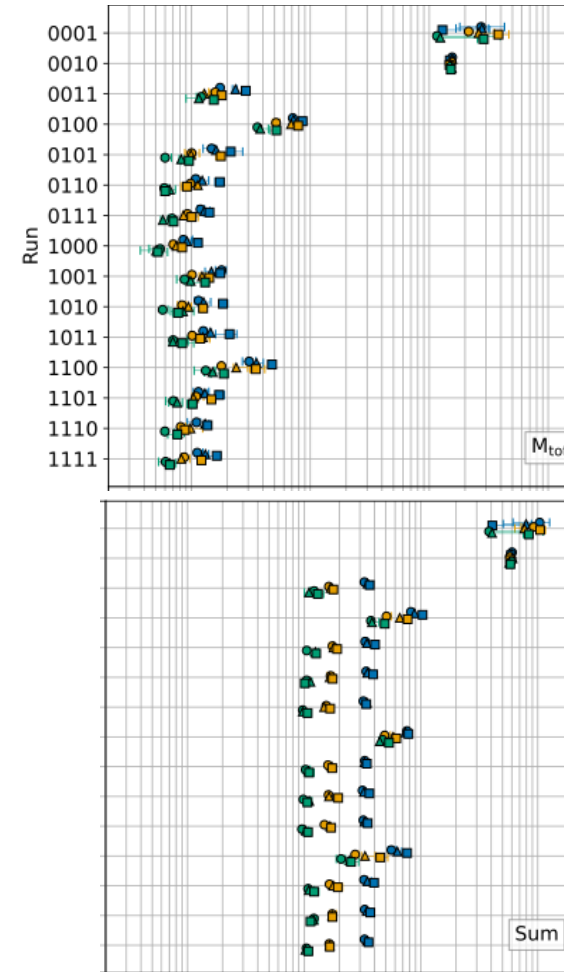
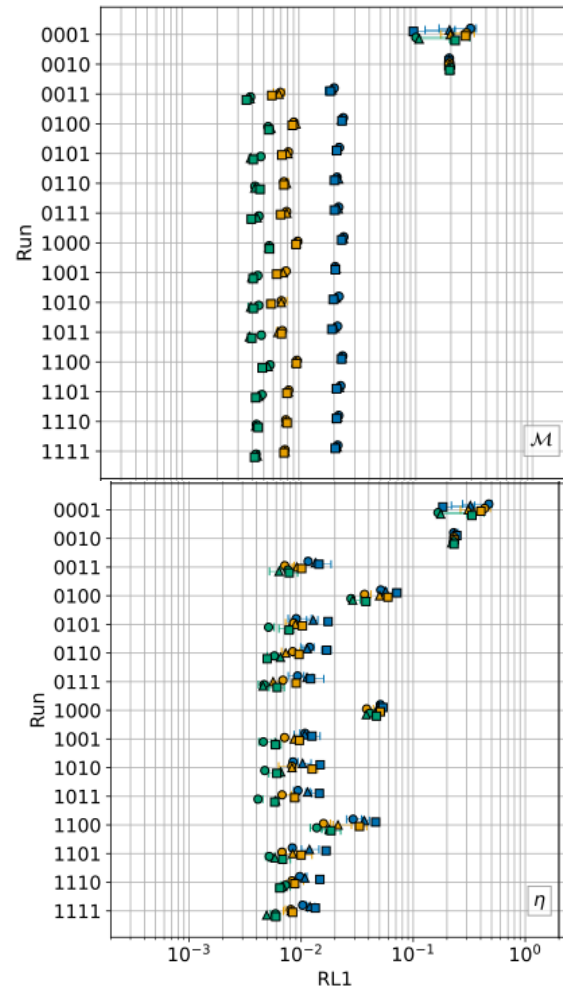
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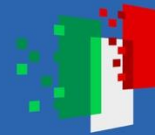


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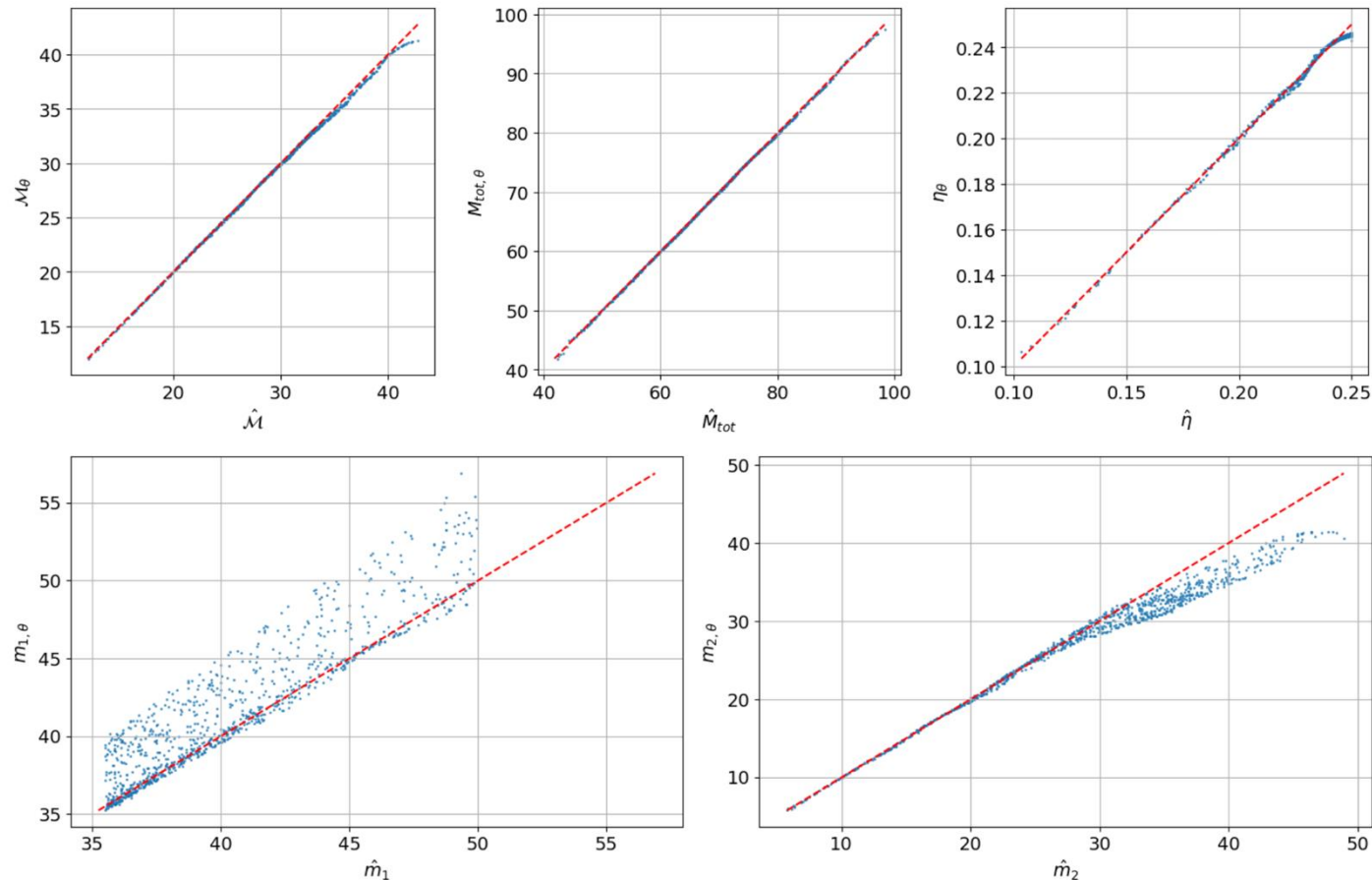


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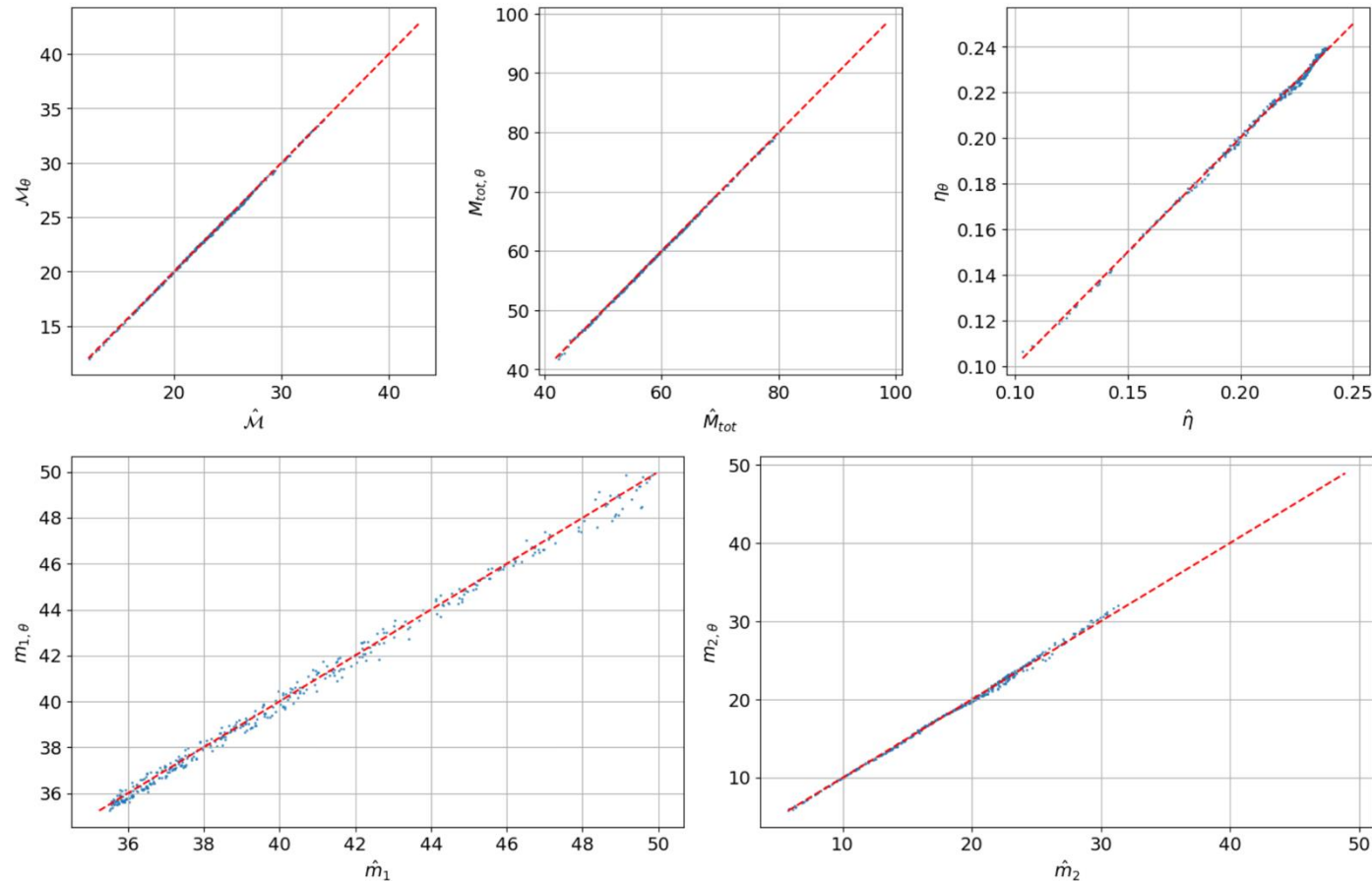


Future steps – fix the borders



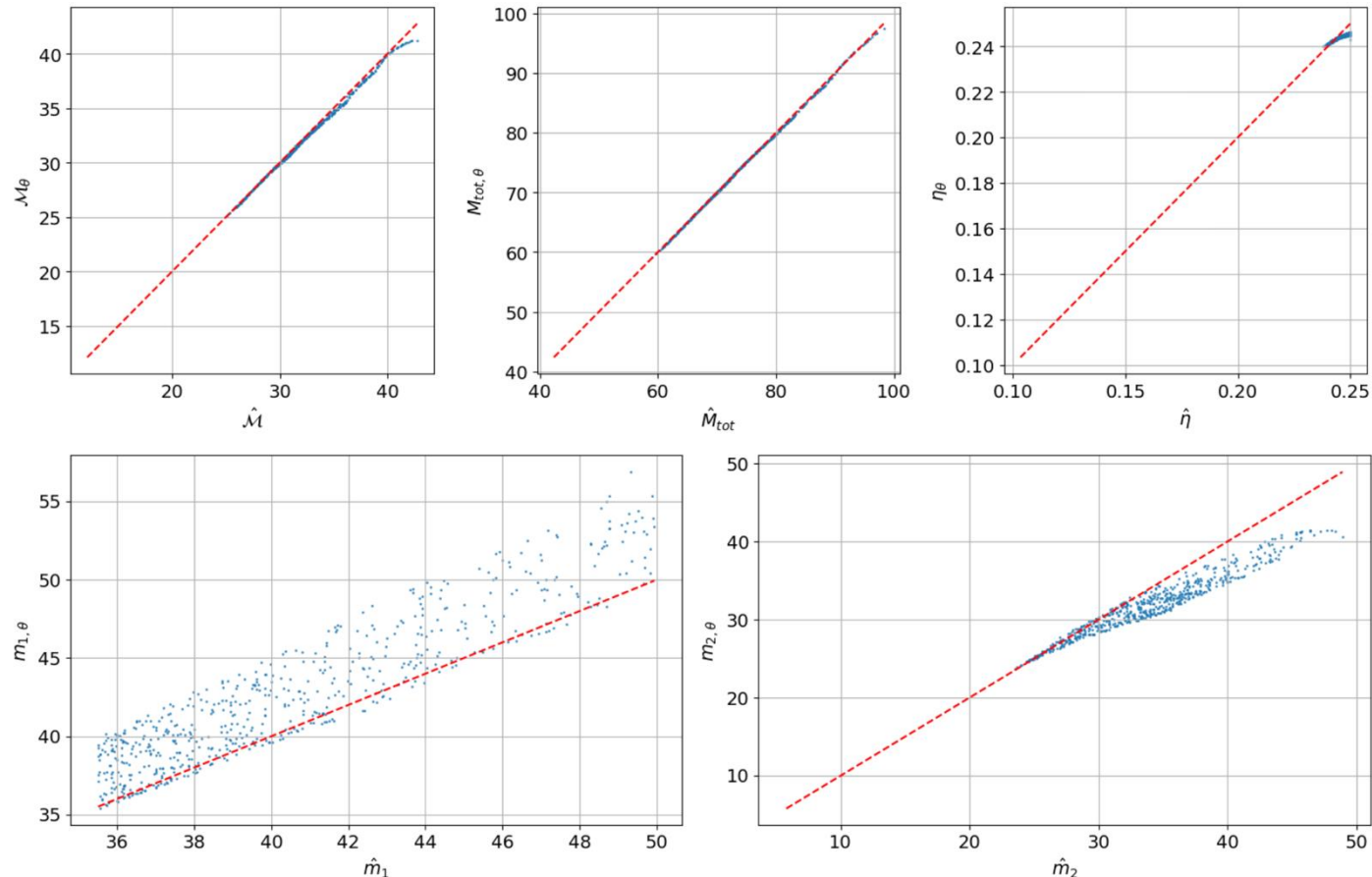


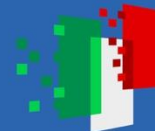
Future steps – fix the borders



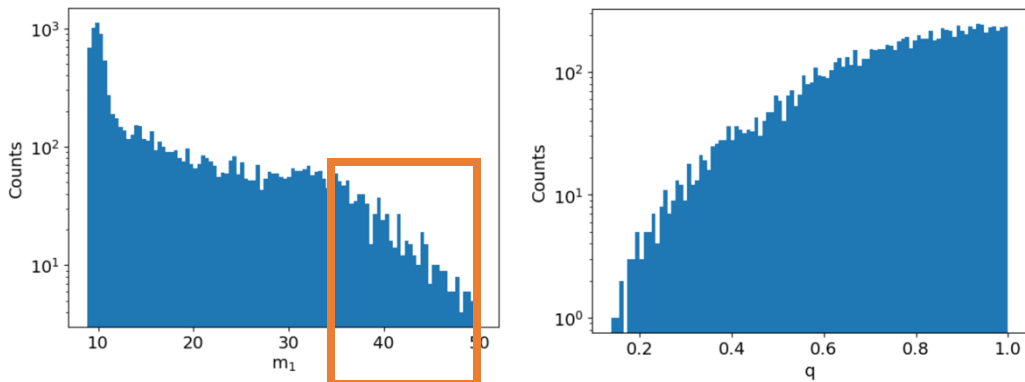


Future steps – fix the borders





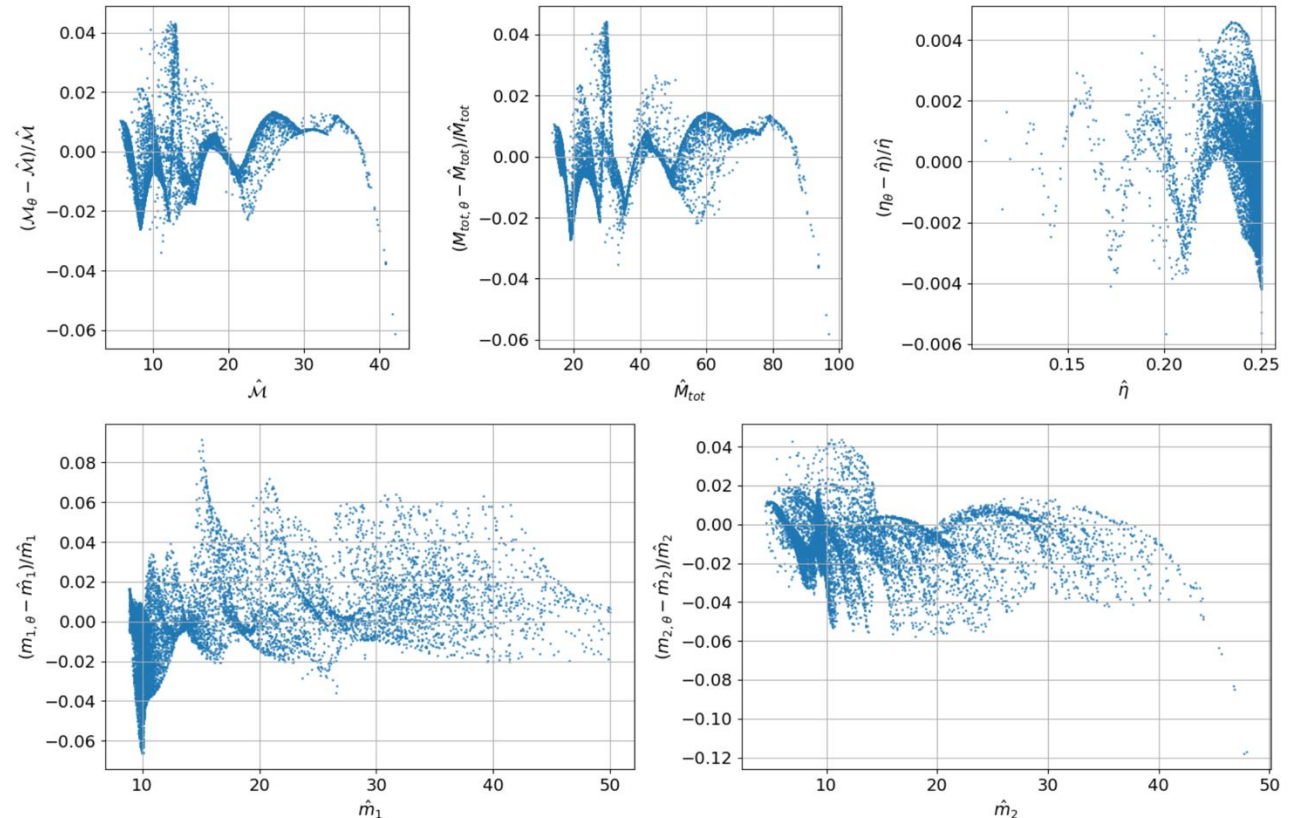
Future steps – Increasing the parameter space to the full GWTC-4 range



Range of APRIL

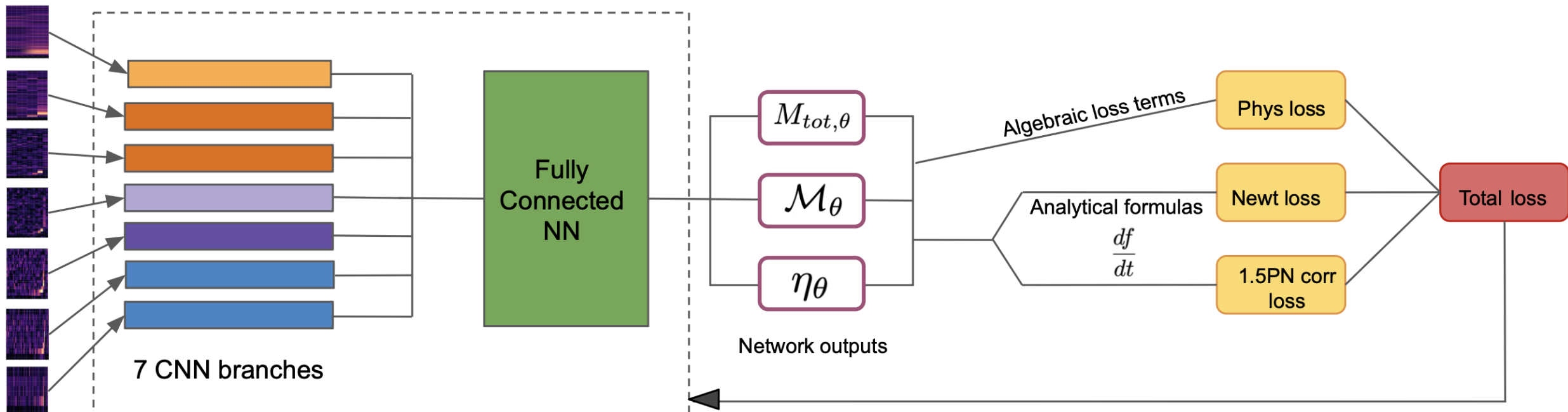
Changes with respect to APRIL:

- extended mass range;
- exp instead of sigmoid as last activation;
- coefficient for eta loss components = 10^8 : best results, but not optimal theoretically.



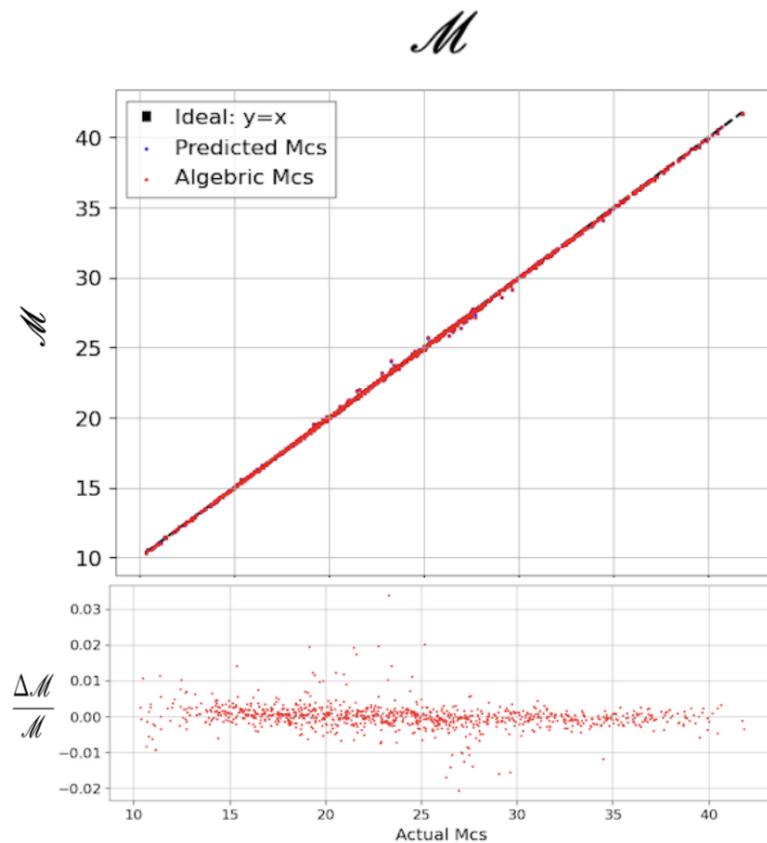


PINNGraPE – CBC PE from cWB outputs

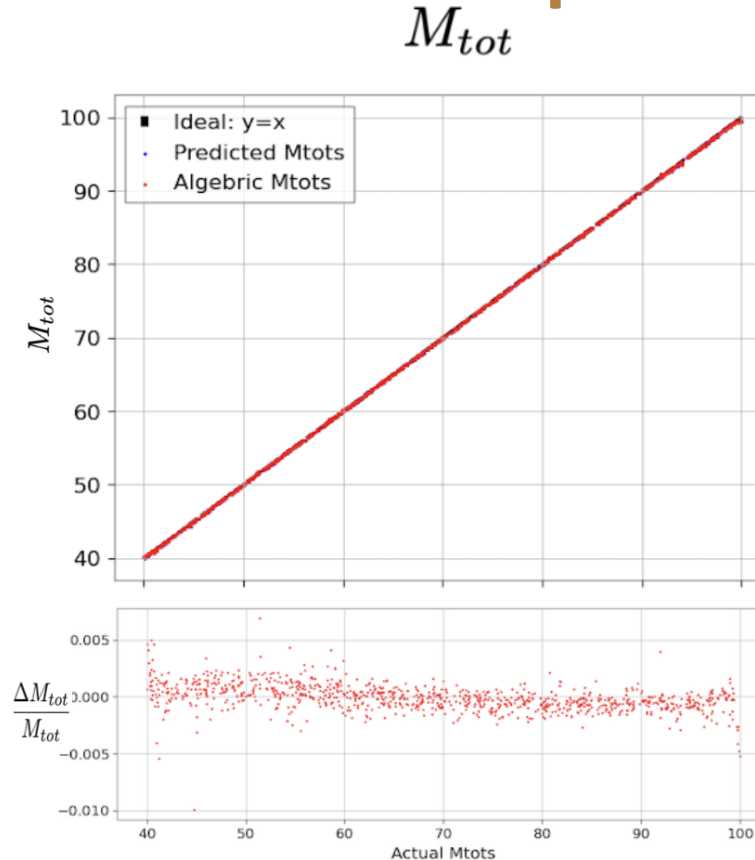




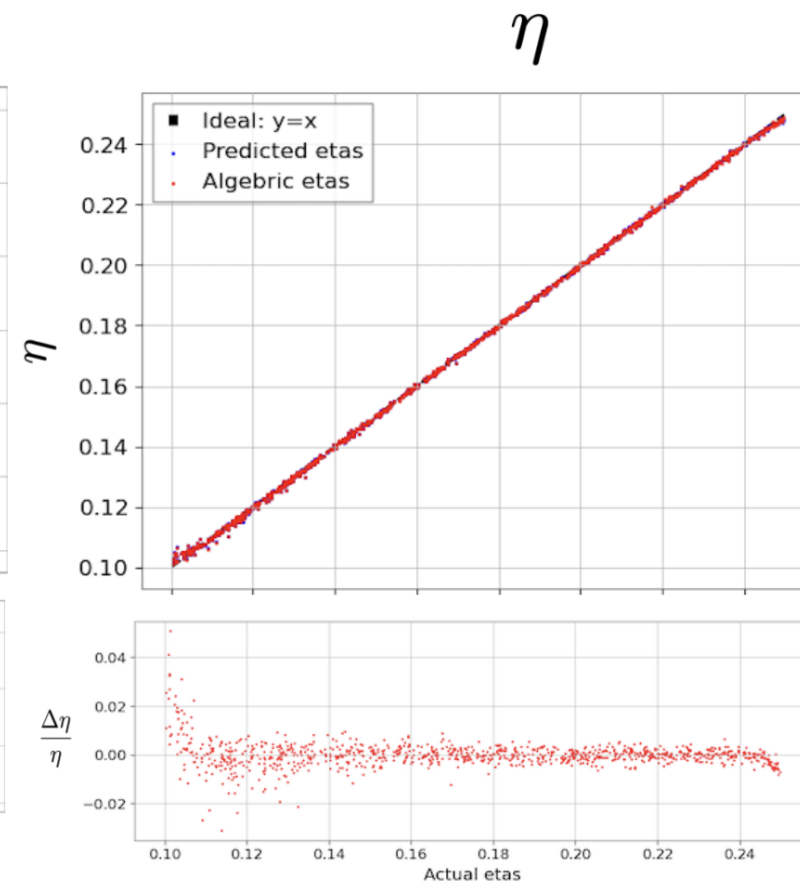
PINNGraPE – CBC PE from cWB outputs



3% relative error



0.5% relative error

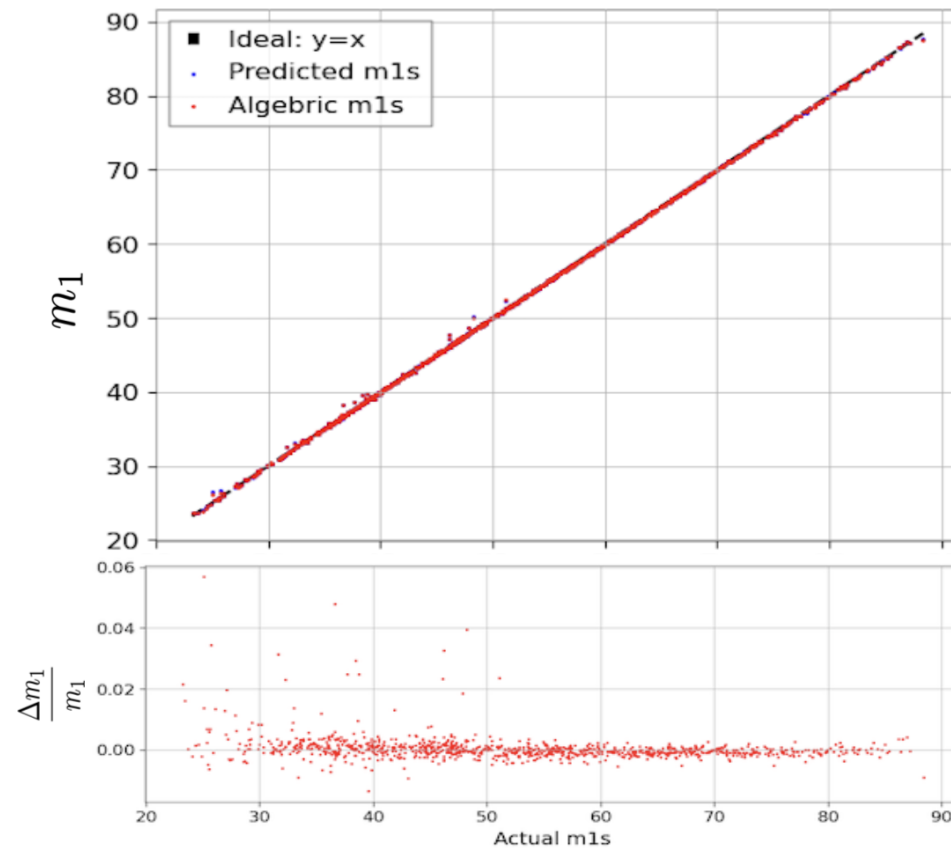


4% relative error



PINNGraPE – CBC PE from cWB outputs

m_1



m_2

m_1 and m_2
inferred
within 6%
error

